

Asynchronous Programming (Async Programming)

Asynchronous programming is a programming model where tasks are executed without blocking the main thread, allowing the program to remain responsive while waiting for operations (like file I/O, network requests, or database queries) to finish.

Instead of stopping the whole program until a task completes, async programming lets the program "do other work" while waiting.

Key Points

- It improves responsiveness in applications (especially UI and web apps).
 - It is not the same as parallel programming:
 - **Async** = handling tasks without blocking.
 - **Parallel** = running tasks at the same time on multiple processors.
 - In C#, it is implemented using the `async` and `await` keywords.
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Simple Analogy

Imagine putting a pizza in the oven:

- **Synchronous programming:** You stand in front of the oven and wait until it's done (doing nothing else).
 - **Asynchronous programming:** You put the pizza in the oven, set a timer, and meanwhile clean the kitchen or scroll your phone. When the timer rings, you take the pizza out.
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Asynchronous programming = writing code that doesn't block execution, allowing programs to stay responsive and efficient while waiting for long-running tasks.